

MASCARA GHRIS, Algeria

WMO: 605070

Lat: 35.208N

Lon: 0.147E

Elev: 514

StdP: 95.30

Time Zone: 1.00 (E01)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.2	1.1	-3.7	2.9	11.1	-2.0	3.4	7.8	14.2	10.9	12.2	10.4	0.6	160	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	40.0	20.8	38.2	20.8	36.8	20.5	23.9	33.8	23.0	32.5	22.2	31.7	5.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.2	16.9	27.5	20.2	15.9	26.5	19.4	15.0	26.2	74.3	33.9	70.5	32.5	67.5	31.8	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	9.0	7.8	DB	-2.6	43.2	1.6	1.0	-3.7	43.9	-4.7	44.6	-5.6	45.1	-6.7	45.9	(4)
(5)			WB	-2.9	25.8	1.7	1.0	-4.1	26.5	-5.1	27.1	-6.0	27.6	-7.3	28.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.9	9.3	9.8	12.5	15.2	19.6	24.3	28.4	28.0	23.6	19.7	13.5	10.5
	DBStd	7.41	2.67	3.16	3.10	3.42	3.67	3.63	2.52	2.51	2.81	3.30	3.34	2.98
	HDD10.0	133	46	37	13	2	0	0	0	0	0	0	6	28
	HDD18.3	1251	281	238	181	105	28	1	0	0	1	22	151	243
	CDD10.0	3026	23	33	91	157	297	430	569	559	409	302	111	44
	CDD18.3	1102	0	0	1	11	67	181	311	301	160	65	6	1
	CDH23.3	13971	1	4	37	235	974	2338	4162	3821	1625	712	56	6
	CDH26.7	7704	0	0	7	68	430	1283	2585	2301	758	263	11	0
Wind	WSAvg	2.7	2.7	3.2	2.8	2.7	2.9	2.8	2.7	2.6	2.5	2.4	2.8	2.5
Precipitation	PrecAvg	351	42	39	36	43	28	11	4	8	20	34	50	36
	PrecMax	558	104	115	172	148	94	73	21	45	57	121	155	84
	PrecMin	190	0	0	0	1	1	1	0	1	0	0	12	0
	PrecStd	91	29	25	36	32	21	17	5	10	17	27	30	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.9	23.1	26.9	31.9	37.0	40.3	42.8	42.1	37.8	34.4	28.2	23.9
		MCWB	12.3	14.3	14.8	18.2	20.4	19.6	20.8	21.1	20.2	17.5	15.0	14.4
	2%	DB	19.2	20.1	23.9	28.3	33.8	37.9	40.2	39.8	35.1	31.9	24.8	20.2
		MCWB	11.4	12.5	14.6	16.6	18.8	20.0	20.7	21.4	19.8	17.5	14.2	12.5
	5%	DB	16.9	18.1	21.6	26.1	31.1	35.8	38.7	38.1	33.1	29.7	22.2	18.1
		MCWB	10.7	11.7	13.9	15.9	17.9	20.0	20.8	21.1	19.5	17.2	13.7	11.8
	10%	DB	15.0	16.2	19.8	23.4	28.9	33.7	37.0	36.2	31.1	27.4	20.0	16.1
		MCWB	10.3	11.0	13.3	15.2	17.2	19.8	20.6	20.8	19.3	17.0	13.3	11.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.4	15.0	17.2	19.4	22.6	24.1	24.8	25.5	23.5	20.8	18.2	14.9
		MCDB	18.9	20.6	22.4	27.1	31.1	33.7	34.8	36.2	31.0	28.3	23.0	21.2
	2%	WB	12.4	13.4	15.9	17.8	20.2	22.3	23.6	24.1	22.3	19.5	16.4	13.5
		MCDB	17.0	18.6	21.6	25.9	29.3	32.9	34.0	34.1	29.3	26.5	21.1	18.4
	5%	WB	11.5	12.4	14.7	16.7	19.1	21.3	22.7	23.1	21.4	18.6	15.1	12.5
		MCDB	15.8	16.9	20.2	24.3	27.5	32.0	33.2	32.5	28.5	25.6	19.9	17.0
	10%	WB	10.6	11.5	13.7	15.6	18.1	20.3	21.9	22.3	20.7	17.8	14.0	11.6
		MCDB	14.3	15.6	18.6	22.0	26.8	30.5	32.9	31.3	27.8	24.7	18.9	15.5
Mean Daily Temperature Range	5% DB	MDBR	11.3	11.3	12.5	13.6	15.2	16.2	17.0	16.4	14.5	13.7	11.1	10.7
		MCDBR	15.4	14.4	16.9	18.7	19.4	18.7	19.0	18.4	17.2	17.0	14.8	14.1
		MCWBR	8.6	8.1	9.2	8.7	7.7	6.3	6.6	6.5	6.3	6.4	7.0	7.8
	5% WB	MCDBR	12.6	12.7	14.9	16.7	16.6	16.7	16.7	16.0	14.4	14.1	12.2	12.2
		MCWBR	7.6	7.7	8.8	8.6	7.8	7.1	6.9	6.5	6.1	6.0	6.6	7.4
Clear-Sky Solar Irradiance	taub	0.305	0.333	0.398	0.419	0.454	0.479	0.535	0.501	0.460	0.402	0.351	0.315	
	taud	2.484	2.401	2.188	2.151	2.078	2.011	1.863	1.956	2.075	2.229	2.379	2.477	
	Ebn at Noon	902	908	868	867	837	814	765	783	796	814	832	865	
	Edh at Noon	84	102	138	150	164	175	202	181	153	120	93	80	
All-Sky Solar Radiation	RadAvg	2.79	3.63	4.89	5.91	6.78	7.62	7.51	6.79	5.40	4.22	2.95	2.50	
	RadStd	0.27	0.37	0.30	0.47	0.46	0.26	0.15	0.24	0.28	0.28	0.23	0.20	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	+0.29	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page